The closure of a generated subsystem is the subsystem itself

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Editorial note. The counted statement is the author's labelled environment at source0.tex lines 358-360 and its proof environment opens at line 361; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. This source is newly approved for this line; the preamble is built from the author's own package lines, macro block and theorem-environment declarations (source lines 4-65), with the author's optioned amsmath line omitted so that it cannot clash with the wrapper's amsmath.

Editorial note. The source carries no label-number dump and the retained counter values are left empty, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label.

Editorial note. Metadata provenance: the source record of the pinned copy and the licence evidence file are both present; the licence is CC BY 4.0 with status supported.

Editorial note. r002 note: the pre-publication gate compares the card's source_version with the source_version recorded in the licence evidence of the same source. This source's evidence file records exactly "2411.13723v3", while the card had carried "arXiv:2411.13723v3 (CC BY 4.0)" copied from the source record; r002 aligns the card with the evidence and changes nothing else. The evidence file itself is not modified.

Editorial scope. One original result of the pinned author manuscript, counted once: the closure of a generated subsystem is the subsystem itself (author label order_closure).

Allowed standard prerequisites (editorial).

- Generated subsystem. The subsystem generated by A is the union of the finite stages obtained by closing under products of two elements already present.. It is the subsystem whose closure is computed.
- Closure operator. The closure under an ordering is a finitary closure operator and distributes over binary unions.. It is the operator applied to the stages.

Lemma 0.1 If $<$ is an HF-ordering of M, $A \subseteq M$ and $\mathrm{cl}_<(A) = A$, then $\mathrm{cl}_<(\langle A\rangle) = \langle A\rangle$.

Proof. Let $\langle A\rangle = \bigcup_{n<\omega} A_n$, where $A_0 = A$ and A_{n+1} is the set of elements of $\langle A\rangle \setminus (A_0 \cup \ldots \cup A_n)$ which are a product of two elements of $A_0 \cup \ldots \cup A_n$. It is enough to prove by induction that, for every n, $\mathrm{cl}_<(A_0 \cup \ldots \cup A_n) = A_0 \cup \ldots \cup A_n$. This is clear for $n = 0$. Assume $a \in A_{n+1} \setminus (A_0 \cup \ldots \cup A_n)$. There are $b, c \in A_0 \cup \ldots \cup A_n$ such that $a = b \cdot c$. If $b > a$ or $c > a$, then $a \in \mathrm{cl}_<(A_0 \cup \ldots \cup A_n) = A_0 \cup \ldots \cup A_n$, a contradiction. Otherwise, $a > b, c$ and $\{b, c\} \subseteq A_0 \cup \ldots \cup A_n$ is the unique pair of elements such that $a > b, c$ and $a = b \cdot c$. □